

Import-Export Linkages as a Channel for Exchange Rate Hedging

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Outline

- 1 Introduction
- 2 Empirical Facts
- 3 Model
- 4 Numerical Analysis
- 5 Conclusion

Motivation

- **Why should I care about exchange rate risk?**
- Firms can hardly predict exchange rates.
- Even harder than predicting typhoons or earthquakes.
- But it is a matter of life and death for many exporting firms!
- For a typical Chinese manufacturing firms, a 5% RMB appreciation could wipe out the profit margins of their products in foreign markets.
- Some firms are more vulnerable to exchange rate shocks, while some are not.
- *Who lives? Who dies? Who tells their stories?*

Background: Exchange Rate Fluctuations

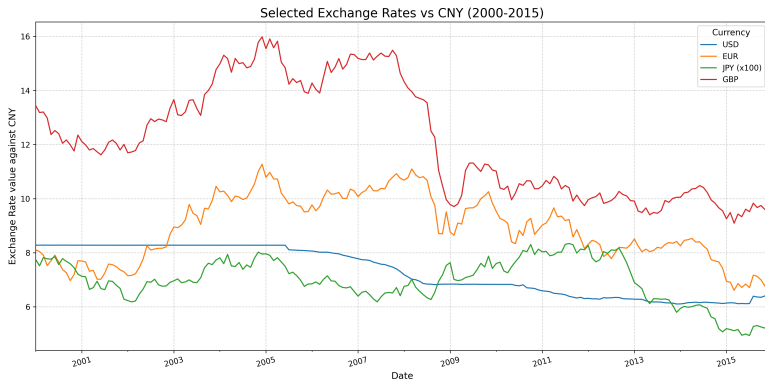


Figure: Nominal exchange rate changes of USD, Euro, JPY and GBP against CNY

- Exchange rate is **volatile**, **persistent**, and **heterogeneous** across currencies.

Background: Export Markets and Import Origins

- Large exporters are often large importers (Amiti, Itskhoki and Konings, 2014).
- Export and import will enhance each other (Li et al., 2024).

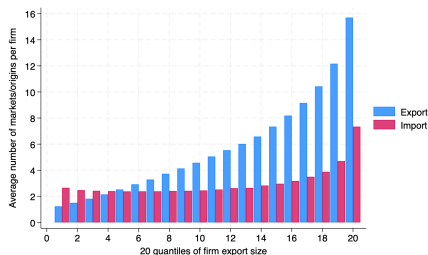


Figure: Large exporters export to more markets and import from more origins

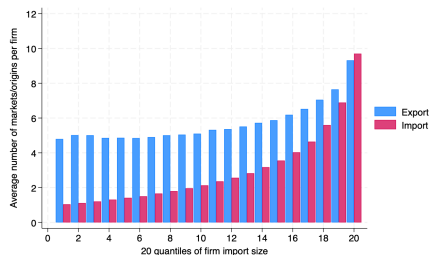


Figure: Large importers import from more origins and export to more markets

► Top Markets and Origins

Risk Hedging with Trade Linkages

- **Problem:** limited financial hedging in emerging markets
 - In China and other emerging markets, limited access to foreign exchange derivatives constrain the financial hedging capabilities of most firms.
 - Exchange rate fluctuations impose persistent uncertainty on firms' export performance $\Rightarrow$ **firms need to hedge in other ways!**
- **Solution: import–export linkages can hedge exchange rate risks.**
 - Conventional trade models either treat exchange rate risk as ex post or only consider complete financial hedging. ► Financial vs Natural
 - I focus on **non-financial** (natural) hedging with trade linkages.
- Opposite effects of exchange rates on trade: **price effect** vs **cost effect**
 - e.g. if there is a home currency appreciation (RMB vs USD)
 - **Price** $\rightarrow$ raise local currency price (or lower producer currency price).
 - **Cost** $\rightarrow$ cheaper imported inputs, lower costs.

Visualization of Multi-country Trade linkages

- Multi-country trade and multilateral exchange rates share a similar structure.

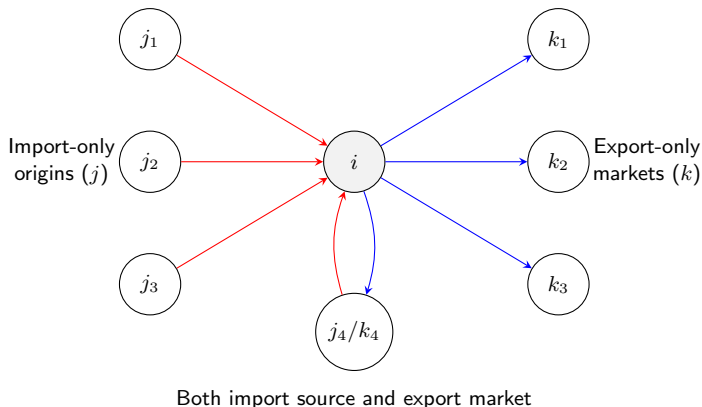


Figure: Multilateral exchange rate shocks in a multi-country trade network

- Exchange rate shocks propagate through this network!

What I Do and Find

- **Empirical:** Three facts from Chinese customs data (2000–2015).
 - ① Two-way traders are less sensitive to bilateral exchange rate shocks.
 - ② A similar network (more overlap) of import origins and export markets reduces sensitivity to bilateral exchange rate shocks.
 - ③ Firm-effective exchange rate (FEER) shocks have opposite effects:
 - Export-weighted FEER $\uparrow \Rightarrow$ export value $\uparrow$;
 - Import-weighted FEER $\uparrow \Rightarrow$ export value $\downarrow$.
- **Model:**
 - Risk-neutral ($\gamma = 0$): existing import and export will cancel off the impact of exchange rate shocks $\rightarrow$ **mechanic** hedging.
 - Risk-averse ($\gamma > 0$): firms will adjust prices and sourcing to reduce profit variance at the cost of expected profit $\rightarrow$ **strategic** hedging.
- **Numerical**
 - Stronger risk aversion increase the import-export alignment for better hedging.
 - Hedging effectiveness depends on the pricing currency, initial import–export alignment, and ER correlation structure.

Connection to Literature

- **Import–export linkages:** Large exporters are large (and diverse) importers; imported inputs shape shock transmission.
 - Amiti, Itskhoki and Konings (2014), Gopinath and Neiman (2014), Halpern, Koren and Szeidl (2015), Blaum (2024), Li et al. (2024).
 - My paper: import-export linkages can reduce exchange rate risk.
- **Hedging of exchange rate risk:**
 - Financial hedging: Allayannis and Ofek (2001), Allayannis and Weston (2001), Salomao and Varela (2022);
 - Non-financial hedging: Raddatz (2011), Fauceglia, Shingal and Wermelinger (2014), Alfaro, Calani and Varela (2021).
 - My paper: proposes a new non-financial hedging channel via trade network.
- **ER pass-through:** Pricing-to-market frameworks connect invoicing currency choices to short-run responses to exchange rate shocks.
 - Gopinath and Rigobon (2008), Gopinath, Itskhoki and Rigobon (2010), Devereux, Dong and Tomlin (2017).
 - My paper: invoicing/pricing currency choice affects natural hedging.

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Data and Measurements

We combine granular trade data and international macro statistics.

- ① Chinese Customs Trade Statistics (CCTS)
 - Source: General Administration of Customs of China
 - Coverage: from 28,000 to 186,000 firms, 2000-2015
- ② International Financial Statistics (IFS) from IMF
 - Exchange rates and inflation; 194 countries since 1948.
- ③ Penn World Table (PWT)
 - Country-level macro variables; 183 countries, 1950-2019

- Bilateral real exchange rates: $RER_{kt} = NER_{kt} \times \frac{CPI_{kt}}{CPI_{it}}$
- Firm effective (value-weighted) exchange rates:

$$\Delta FEER_{ft}^X = \sum_i \frac{V_{fkt}^X}{\sum_{k' \in \mathcal{K}} V_{fk't}^X} \Delta RER_{kt}, \quad \Delta FEER_{ft}^M = \sum_j \frac{V_{fjt}^M}{\sum_{j' \in \mathcal{J}} V_{fj't}^M} \Delta RER_{jt}$$

Empirical Strategy

- **Question:** Does the export response of firms to exchange rate shocks depend on import-export linkages?
- The baseline specifications:

$$\Delta V_{fkt}^X = \beta_1 \Delta RER_{kt} + \beta_2 \Delta RER_{kt} \times \tilde{s}_{f(k)t-1}^M + \gamma Z_{ft-1} + \eta G_{kt} + \delta_f + \epsilon_t \quad (1)$$

$$\Delta V_{ft}^X = \beta_1 \Delta FEER_{ft}^X + \beta_2 \Delta FEER_{ft}^M + \gamma Z_{ft-1} + \delta_f + \epsilon_t \quad (2)$$

- V_{fkt}^X : firm-country-level export value change;
- $\tilde{s}_{f(k)t-1}^M$: trade linkage terms, including two-way indicators $TW_{f(k)t-1}$, lag import intensity $s_{f(k)t-1}^M$ and import-export similarity $Jaccard_{ft}$ and ϕ_{ft} .
- Z_{ft-1} : a vector of firm-level lagged control variables;
- G_{kt} : country-level gravity controls;
- δ_f : firm-level (time-invariant) fixed effects;

Fact 1: Firm-level Two-way Traders

Table: Firm-level Two-way Linkages and Export Value Elasticity

Sample Dependent Var	(1) All Exporters	(2) $\Delta \ln V_{fkt}^X$	(3) Two-way Traders	(4)
$\Delta \ln RER_{kt}$	0.502*** (0.075)	0.651*** (0.077)	0.511*** (0.092)	0.623*** (0.118)
$\Delta \ln RER_{kt} \times TW_{ft-1}$	-0.533*** (0.117)	-0.747*** (0.121)		
$\Delta \ln RER_{kt} \times s_{ft-1}^M$			-0.963*** (0.303)	-1.318*** (0.367)
$\Delta \ln RGDP_{kt}$	1.358*** (0.210)	2.076*** (0.266)	1.440*** (0.246)	2.148*** (0.312)
Lag Firm Controls	Yes	Yes	Yes	Yes
Gravity Controls	Yes	No	Yes	No
Firm-country FE	No	Yes	No	Yes
Firm FE	Yes	No	Yes	No
Year FE	Yes	Yes	Yes	Yes
Observations	5019959	4375637	1492001	1305681
R^2	0.117	0.196	0.102	0.187

Notes: The dependent variables are firm-country-level export value changes. *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

- Two-way traders are on average less sensitive to bilateral RER shocks.

Fact 1: Country-specific Two-way Traders

Table: Country-specific Two-way Linkages and Export Value Elasticity

Sample Dependent Var	(1)	(2) Two-way Traders $\Delta \ln V_{fkt}^X$	(3)	(4)
$\Delta \ln RER_{kt}$	0.450*** (0.079)	0.502*** (0.084)	0.938*** (0.109)	1.229*** (0.162)
$\Delta \ln RER_{kt} \times TW_{fkt-1}$	-0.472*** (0.106)	-0.512*** (0.127)		
$\Delta \ln RER_{kt} \times s_{fkt-1}^M$			-4.440*** (0.574)	-5.700*** (0.802)
$\Delta \ln RGDP_{kt}$	1.423*** (0.243)	2.102*** (0.310)	1.322*** (0.248)	1.882*** (0.319)
Lag Firm Controls	Yes	Yes	Yes	Yes
Gravity Controls	Yes	No	Yes	No
Firm FE	Yes	No	Yes	No
Firm-country FE	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1492001	1305681	1492001	1305681
R^2	0.102	0.187	0.105	0.190

Notes: The dependent variables are firm-country-level export value changes. *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

- Two-way traders import from the exactly same country it exports are even less sensitive to its bilateral RER shocks.

Fact 2: Network Similarity

Table: Firm-level Network Similarity and Export Value Elasticity

Sample Dependent Var	(1)	(2) Two-way Traders $\Delta \ln V_{fkt}^X$	(3)	(4)
$\Delta \ln RER_{kt}$	0.420*** (0.125)	0.484*** (0.125)	0.541*** (0.161)	0.651*** (0.163)
$\Delta \ln RER_{kt} \times Jaccard_{ft-1}$	-0.536* (0.308)	-0.673** (0.315)		
$\Delta \ln RER_{kt} \times \phi_{ft-1}$			-0.594** (0.298)	-0.777** (0.315)
$\Delta \ln RGDP_{kt}$	1.435*** (0.248)	2.138*** (0.315)	1.435*** (0.248)	2.138*** (0.315)
Lag Firm Controls	Yes	Yes	Yes	Yes
Gravity Controls	Yes	No	Yes	No
Firm FE	Yes	No	Yes	No
Firm-country FE	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1492001	1305681	1492001	1305681
R^2	0.102	0.187	0.102	0.187

Notes: The dependent variables are firm-country-level export value changes. *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

- Two-way traders with similar import and export country sets are significantly less sensitive to bilateral RER shocks.

Fact 3: Firm-Effective Exchange Rates

Table: Firm Effective Exchange Rates and Export Value Elasticity

Sample Dependent Var	(1) All Exporters	(2)	(3) Two-way Traders $\Delta \ln V_{ft}^X$	(4)
$\Delta \ln FEER_{ft}^X$	0.165*** (0.029)	0.036 (0.057)		0.108* (0.063)
$\Delta \ln FEER_{ft}^M$			-0.100** (0.047)	-0.144*** (0.052)
Lag Firm Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1285518	344163	344163	344163

Notes: The dependent variables are firm-level export value changes. Column includes all exporters. Column (2)-(4) include two-way traders who both export and import. *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level and country-year level.

- Value-weighted firm effective exchange rate shocks have opposite effects:
 - Export FEER is **positively** correlated with export value;
 - Import FEER is **negatively** correlated with export value;
 - Export and import FEERs can offset each other.

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 - Risk-averse Extension
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Environment & Technology

- N countries indexed by i (host), j (import origin) and k (export destination)
- Consumer utility in the CES objective

$$U_i = \left(\int_{\omega \in \Omega_i} q_i(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \sigma > 1 \quad (3)$$

- Cobb-Douglas production uses labor L , material inputs X and technology φ

$$y_i(\omega) = \varphi(\omega) L_i^{1-\alpha} X_i^\alpha \quad (4)$$

- Input bundle nests a two-layer CES between domestic and imported inputs:

$$X_i = \left[X_{D,i}^{\frac{\kappa-1}{\kappa}} + X_{M,i}^{\frac{\kappa-1}{\kappa}} \right]^{\frac{\kappa}{\kappa-1}}, \quad X_{M,i} = \left(\sum_{j \in \mathcal{J}} X_{ij}^{\frac{\kappa-1}{\kappa}} dj \right)^{\frac{\kappa}{\kappa-1}}, \kappa > 1 \quad (5)$$

- Unit production cost under constant return to scale is:

$$c_i(\omega) = \varphi(\omega)^{-1} \alpha^{-\alpha} (1-\alpha)^{\alpha-1} w_i^{1-\alpha} P_{X,i}(\omega)^\alpha \quad (6)$$

Firm's Export Pricing Decision

- Dixit–Stiglitz demand in both domestic market i and foreign markets k .
- The export revenue in market k is a mix of producer currency pricing (PCP) and local currency pricing (LCP) by parameter λ :

$$R_{ki} = \lambda R_{ki}^{PCP} + (1 - \lambda) R_{ki}^{LCP} = \lambda e_{ik}^{\sigma} B_{ki}^{PCP} + (1 - \lambda) e_{ik} B_{ki}^{LCP} \quad (7)$$

where $B_{ki}^{PCP} \equiv (\frac{\bar{P}_{ki}^i}{P_k})^{1-\sigma} \frac{S_k}{P_k}$ and $B_{ki}^{LCP} \equiv (\frac{\bar{P}_{ki}^k}{P_k})^{1-\sigma} \frac{S_k}{P_k}$ are price anchors independent of exchange rates.

- The optimal export share to country k is:

$$\chi_{ki}^X \equiv \frac{R_{ki}}{R_i^X} = \frac{\lambda e_{ik}^{\sigma} B_{ki}^{PCP} + (1 - \lambda) e_{ik} B_{ki}^{LCP}}{\sum_{k'} [\lambda e_{ik'}^{\sigma} B_{k'i}^{PCP} + (1 - \lambda) e_{ik'} B_{k'i}^{LCP}]} \quad (8)$$

Firm's Import Sourcing Decision

- The intermediate input price index P_X :

$$P_{X,i} = (P_{D,i}^{1-\epsilon} + P_{M,i}^{1-\epsilon})^{\frac{1}{1-\epsilon}}, \quad P_{M,i} = \left(\sum_{j \in \mathcal{J}} p_{ij}^{1-\kappa} dj \right)^{\frac{1}{1-\kappa}} \quad (9)$$

- Assume import price is subject to complete PCP: $p_{ij} = p_{ij}^* e_{ij}$.
- Import share from country j over total imported input:

$$\chi_{ij}^M = \frac{p_{ij} \left(\frac{p_{ij}}{P_{M,i}} \right)^{-\kappa}}{\sum_{j' \in \mathcal{J}} p_{ij'} \left(\frac{p_{ij'}}{P_{M,i}} \right)^{-\kappa}} = \frac{(p_{ij}^* e_{ij})^{1-\kappa}}{\sum_{j' \in \mathcal{J}} (p_{ij'}^* e_{ij'})^{1-\kappa}} \quad (10)$$

- Overall input price $P_X = P_D [1 + A^{\epsilon-1}]^{\frac{1}{1-\epsilon}}$, $A_i = P_{D,i}/P_{M,i}$
 - Firms enjoy cost advantage from cheaper imported inputs.
- The optimal share of input expenditure on imported inputs (import intensity):

$$s_i^M = \frac{A_i^{\epsilon-1}}{1 + A_i^{\epsilon-1}}, \quad 0 < s_i^M < 1 \quad (11)$$

Profit Maximization in Static Partial Equilibrium

- The firm ω 's profit is given as:

$$\pi_i(\omega) = p_{ii}(\omega)q_{ii}(\omega) + \sum_{\mathcal{K}} \frac{p_{ki}(\omega)q_{ki}(\omega)}{e_{ik}} - w_i L - P_{D,i} X_{D,i} - P_{M,i} X_{M,i} - F(\mathcal{J}, \mathcal{K}) \quad (12)$$

$$\text{s.t. } q_{ii}(\omega) + \tau_{ki} q_{ki}(\omega) = Y_i(\omega) = \varphi(\omega) L_i^{1-\alpha} X_i^\alpha.$$

- In a static partial equilibrium:
 - Exogenous: firm productivity φ , foreign price P_k , domestic and foreign demand S_i and S_k ; trade partner set $\mathcal{J}, \mathcal{K} \subseteq N$, labor L ;
 - Firms make **intensive margin decisions**: intermediate input bundle $X_{D,i}$ and $X_{M,i}$, local price in domestic and foreign markets p_{ii} and p_{ki} .
 - Facing exchange rate shocks, solve the **local adjustments** of import/export intensity s_i^M and s_i^X , and import/export shares of each country χ_{ij}^M and χ_{ki}^X .

Static Exchange Rate Elasticities

- Exchange elasticities of cost c_i :

$$\varepsilon(c_i, e_{il}) = \alpha s_i^M \chi_{il}^M > 0 \quad (13)$$

- Exchange elasticities of export revenue R_i^X :

$$\varepsilon(R_i^X, e_{il}) = \underbrace{[\lambda\sigma + 1 - \lambda]\chi_{li}^X}_{\text{price effect}(+)} + \underbrace{(1 - \sigma)\alpha s_i^M \chi_{il}^M}_{\text{cost effect}(-)} \quad (14)$$

- Exchange rate elasticity of import intensity s_i^M and export intensity s_i^X :

$$\varepsilon(s_i^M, e_{il}) = (1 - \epsilon)(1 - s_i^M)\chi_{il}^M < 0 \quad (15)$$

$$\varepsilon(s_i^X, e_{il}) = \sigma(1 - s_i^X)\chi_{li}^X > 0 \quad (16)$$

- Exchange rate elasticity of import share χ_{ij}^M , and export share χ_{ki}^X :

$$\varepsilon(\chi_{ij}^M, e_{il}) = (1 - \kappa)(\mathbf{1}\{j = l\} - \chi_{il}^M) \rightarrow \text{direct} < 0, \text{ indirect} > 0 \quad (17)$$

$$\varepsilon(\chi_{ki}^X, e_{il}) = (\lambda\sigma + 1 - \lambda)(\mathbf{1}\{k = l\} - \chi_{li}^X) \rightarrow \text{direct} > 0, \text{ indirect} < 0 \quad (18)$$

Profit Exposure Decomposition

- Exchange rates follow random walk with covariance: $\varepsilon_t^e \sim \mathcal{N}(\mathbf{0}, \Sigma^e)$
- Decompose profit deviation into revenue and cost deviations ($\tilde{x} = x - \mathbb{E}[x]$):

$$\tilde{\pi}_i = \tilde{R}_{ii} + \sum_k \tilde{R}_{ki} - \tilde{C}_{M,i} \quad (19)$$

- Decompose profit deviation into exposures to bilateral exchange rate shocks:

$$\tilde{\pi}_i = \sum_l u_{il} \Delta e_{il} \quad (20)$$

→ Each bilateral exposure can be decomposed into three parts

$$u_{il} = \underbrace{a_{il}}_{\text{domestic sales}} + \sum_k \underbrace{b_{il,k}}_{\text{export sales}} - \underbrace{c_{il}}_{\text{import costs}} \quad (21)$$

- $\mathbf{u}_i$ encodes multilateral exposure shaped by structural parameters.

$$\mathbf{u}_i = (u_{i1}, u_{i2}, \dots, u_{iN})^\top = (a_{i1} + \sum_k b_{i1}^k - c_{i1}, \dots, a_{iN} + \sum_k b_{iN}^k - c_{iN})^\top \quad (22)$$

Optimization under Risk Aversion

- Constant absolute risk aversion (CARA) with parameter $\gamma > 0$:

$$J = \underbrace{\mathbb{E}[\pi_i]}_{\text{Expected Profit}} - \underbrace{\frac{\gamma}{2} \cdot \text{Var}(\pi_i)}_{\text{Risk Loss}} \quad (23)$$

- Profit variance in matrix form: $\text{Var}(\pi_i) = \mathbf{u}_i^\top \Sigma^e \mathbf{u}_i$.
- First-order conditions (FOCs) of intensive margin decisions:

$$\frac{\partial \mathbb{E}[\pi_i]}{\partial \theta} = \gamma \left(\frac{\partial \mathbf{u}_i}{\partial \theta} \right)^\top \Sigma^e \mathbf{u}_i \quad (24)$$

- What happen to firms' intensive margin decision if they are risk averse?
 - ① export price premium: export more or less by charging a different markup
 - ② import sourcing shift: import more or less by applying a "shadow unit cost".
- **More risk aversion $\rightarrow$ Further tradedeviations $\rightarrow$ More distortions**

Risk-Adjusted Pricing Equations

- FOCs of domestic price p_{ii} and export price p_{ki} :

$$\frac{\partial \mathbb{E}[\pi_i]}{\partial p_{ii}} - \frac{\gamma}{2} \frac{\partial \text{Var}(\pi_i)}{\partial p_{ii}} = 0, \quad \frac{\partial \mathbb{E}[\pi_i]}{\partial p_{ki}} - \frac{\gamma}{2} \frac{\partial \text{Var}(\pi_i)}{\partial p_{ki}} = 0 \quad (25)$$

- The domestic risk premium ψ_{ii} and the export risk premium ψ_{ki} :

$$\psi_{ii} = \gamma(1 - \sigma)\alpha s_i^M \sum_{j,l} \chi_{ij}^M u_{il} \sigma_{jl}^e, \quad \psi_{ki} = \gamma \sum_{j,l} \varepsilon_{ij}^{R_k} u_{il} \sigma_{jl}^e \quad (26)$$

- Rewrite domestic and export risk premia with the profit exposure factor $\mathbf{u}_i$:

$$\underbrace{\psi_{ii} = \gamma \phi_i^M (\bar{\chi}_i^M)^\top (\Sigma^e \mathbf{u}_i)}_{\text{cost effect (-)}}, \quad \psi_{ki} = \underbrace{\psi_{ii} + \gamma \phi_i^X (\Sigma^e \mathbf{u}_i)_k}_{\text{price effect (+)}} \quad (27)$$

where $\phi_i^M = (1 - \sigma)\alpha s_i^M < 0$ and $\phi_i^X = \lambda\sigma + 1 - \lambda > 0$ are sensitivity factors.

Risk-Adjusted Import Allocation

- FOC of import shares under risk aversion:

$$\frac{\partial \mathbb{E}[C_{M,i}]}{\partial \chi_{ij}^M} + \gamma(\mathbf{d}_{ij}^M)^\top \Sigma^e \mathbf{u}_i = \lambda_M \quad (28)$$

- $d_{ij}^M \equiv \frac{\partial u_i^C}{\partial \chi_{ij}^M} \in \mathbb{R}^J$ is the marginal impact of import shares on cost exposure.
- The “effective unit sourcing cost” or “shadow price”:

$$\tilde{p}_{ij} \equiv p_{ij} + \Gamma_M \gamma(\mathbf{d}_{ij}^M)^\top \Sigma^e \mathbf{u}_i, \quad \Gamma^M > 0 \quad (29)$$

- $\tilde{p}_{ij} > p_{ij}$: import from j increase net risk exposure $\rightarrow$ “escape”.
- $\tilde{p}_{ij} < p_{ij}$: import from j decrease net risk exposure $\rightarrow$ “buy more”.
- The import shares under risk aversion deviates from the risk-neutral shares:

$$\chi_{ij}^M = \frac{(\tilde{p}_{ij})^{1-\kappa}}{\sum_{j' \in \mathcal{J}} (\tilde{p}_{ij'})^{1-\kappa}} = \frac{(p_{ij} + \Gamma_M \gamma(\mathbf{d}_{ij}^M)^\top \Sigma^e \mathbf{u}_i)^{1-\kappa}}{\sum_{j'} (p_{ij'} + \Gamma_M \gamma(\mathbf{d}_{ij'}^M)^\top \Sigma^e \mathbf{u}_i)^{1-\kappa}} \quad (30)$$

Linear Equation System under Risk Aversion

- The risk-averse pricing equation as:

$$A_p \Delta z + \gamma R_p \Sigma^e \Delta u = B_p \Delta s + C_p \Psi \quad (31)$$

- Δs : exogenous exchange rate shocks.
- Δz : endogenous variables.
- Ψ : level factors and other unmodeled shifts.
- R_p : risk-loading block (sensitivity to exposure changes)
- Net risk exposure change Δu absorbs export premium and import allocation:

$$\Delta u = T_u(\gamma)(N_z \Delta z + N_s \Delta s + N_\Psi \Psi) \quad (32)$$

- N_z , N_s , and N_Ψ : structural coefficients of exposure adjustments.
- The final linear equation system with risk aversion:

$$\underbrace{(A_z + \gamma R_p \Sigma^e T_u N_z)}_{A^*(\gamma)} \Delta z = \underbrace{(B_p - \gamma R_p \Sigma^e T_u N_s)}_{B^*(\gamma)} \Delta s + \underbrace{(C_p - \gamma R_p \Sigma^e T_u N_\psi)}_{C^*(\gamma)} \psi \quad (33)$$

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Numerical Overview

- Why analytical solutions are not enough?
- The story is not finished yet, and I need to:
 - evaluate the effectiveness of natural hedging;
 - consider the heterogeneity of firms;
 - conduct some simple counterfactual experiments.
- Monte-Carlo simulation can help!
 - I simulate 10000 exchange rate shocks $\Delta e \sim \mathcal{N}(0, \hat{\Sigma})$ of 20 currencies.
 - Average initial import and export intensities and shares are from customs data.
 - The objective is to observe the hedging effectiveness under simulated shocks.

► Parametrization

Metrics of Hedging Effectiveness

- Exposure-offset share (EOS)**

$$\text{EOS}_i = 1 - \frac{|\tilde{R}_i - \tilde{C}_i|}{|\tilde{R}_i| + |\tilde{C}_i|} = 1 - \frac{\|\mathbf{u}_i\|}{\|\mathbf{u}_i^R\| + \|\mathbf{u}_i^C\|} \quad (34)$$

- Variance-reduction ratio (VRR)**

$$\text{VRR}_i = \frac{2\text{Cov}(\tilde{R}_i, \tilde{C}_i)}{\text{Var}(\tilde{R}_i) + \text{Var}(\tilde{C}_i)} = \frac{2(\mathbf{u}_i^R)^\top \Sigma^e (\mathbf{u}_i^C)}{(\mathbf{u}_i^R)^\top \Sigma^e \mathbf{u}_i^R + (\mathbf{u}_i^C)^\top \Sigma^e \mathbf{u}_i^C} \in [-1, 1] \quad (35)$$

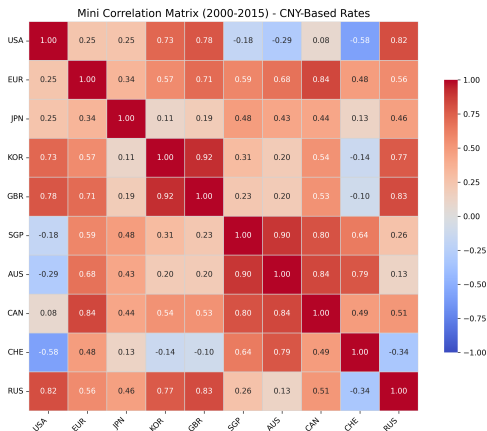
- Revenue-cost correlation (ρ^{RC})**

$$\rho_i^{RC} = \frac{\text{Cov}(\tilde{R}_i, \tilde{C}_i)}{\sqrt{\text{Var}(\tilde{R}_i)\text{Var}(\tilde{C}_i)}} = \frac{\mathbf{u}_i^R{}^\top \Sigma^e \mathbf{u}_i^C}{\sqrt{[(\mathbf{u}_i^R)^\top \Sigma^e \mathbf{u}_i^R][(\mathbf{u}_i^C)^\top \Sigma^e \mathbf{u}_i^C]}} \in [0, 1] \quad (36)$$

- Higher EOS/VRR/ $\rho^{RC} \rightarrow$ Stronger hedging through trade linkages.**

Estimated Covariance Matrix

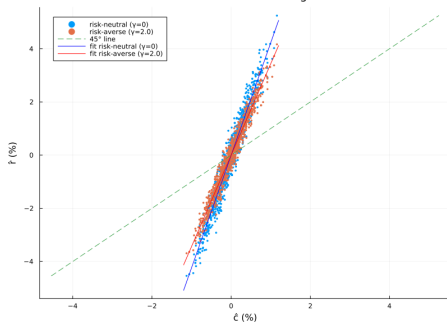
- Sample covariance matrix using historical data: $\hat{\Sigma}_S = \frac{1}{T-1} \sum_{t=1}^T X_t X_t'$
 - $X_t = r_t - \frac{1}{T} \sum_{t=1}^T r_t$: observed deviation from sample mean of log returns r_t .
 - Ledoit-Wolf shrinkage $\hat{\Sigma}_{LW} = \delta F_{cc} + (1 - \delta) \hat{\Sigma}_S$ [Full Covariance Matrix](#)



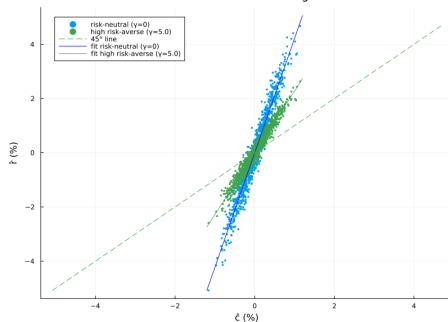
Monte-Carlo Simulation: Revenue vs Cost

- Revenue and cost changes are closer to the 45-degree line with higher γ .

Revenue vs Cost Changes

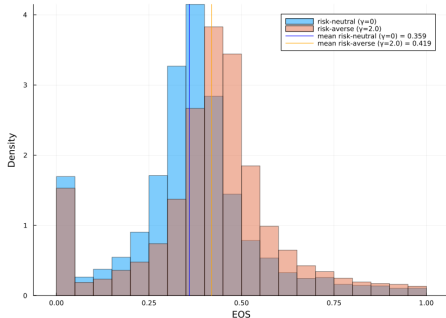


Revenue vs Cost Changes

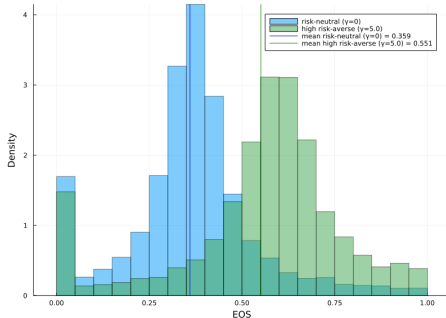


Monte-Carlo Simulation: EOS Distribution

Exposure-Offset Shares (EOS)

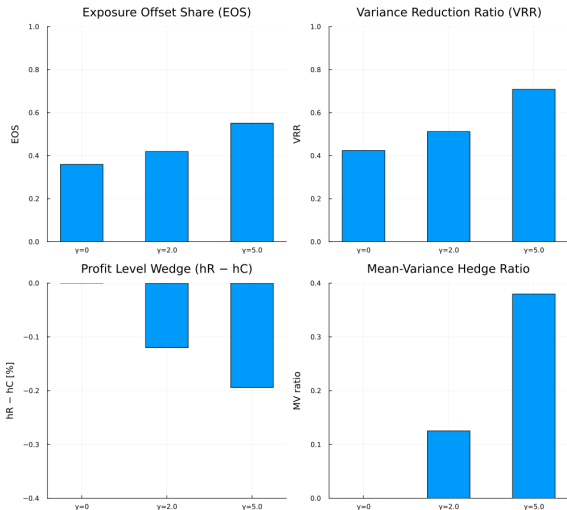


Exposure-Offset Shares (EOS)



- The distribution of EOS shifts to the right with higher risk aversion γ .

Average Effectiveness and Cost of Natural Hedging



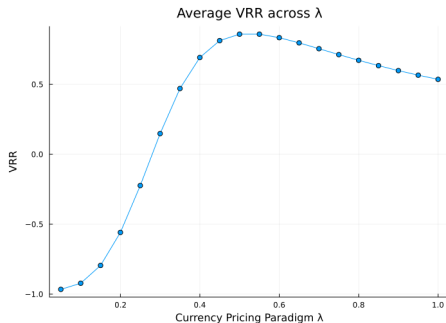
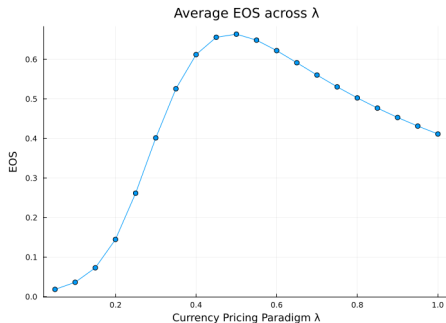
Higher risk aversion γ

- Net risk exposure ↓
- Profit variance ↓
- Expected profit ↓
- **MV trade-off** ↑

Figure: Effectiveness and Cost of Natural Hedging

Export Pricing Scheme: PCP vs LCP

- Export pricing paradigm $\lambda \in [0, 1]$ between PCP ($\lambda = 1$) and LCP ($\lambda = 0$)



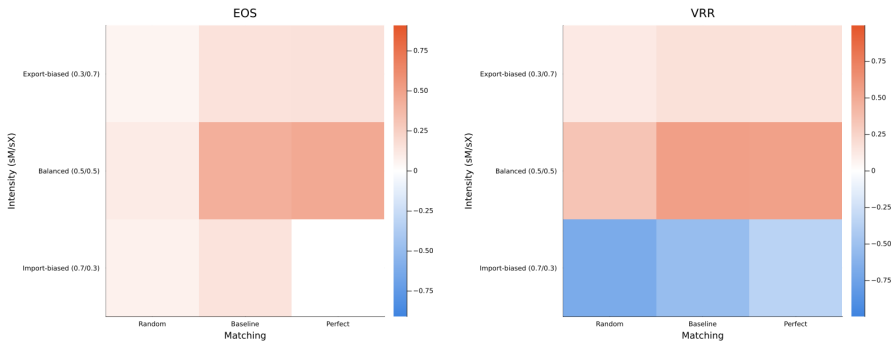
- Hump-shape $\rightarrow$ Best average hedging in a mixed PCP/LCP case $\lambda \approx 0.5$.

Overview of Counterfactual Experiments

- Alternative initial trade structure
 - Balance of import and export intensities: import-biased ($s^M > s^X$), balanced ($s^M \approx s^X$), export-biased ($s^M < s^X$).
 - Matching of import and export shares: random (no correlation $\vec{\chi}^M$ and $\vec{\chi}^X$), baseline ($\vec{\chi}_0^M$ and $\vec{\chi}_0^X$ from data), perfect ($\vec{\chi}^M = \vec{\chi}^X$).
- Hypothetical exchange rate environments
 - Currency peg: fixed exchange rates with USD and the 20-currency basket.
 - Volatility amplification: global hikes (all elements in $\Sigma^e \times 2$) and idiosyncratic hikes (only diagonal elements in $\Sigma^e \times 2$).
 - Jump-mix risk: USD and EUR ($p = 10\%$ and $|J| = 20\%$). [▶ Jump-mix](#)

Counterfactual Initial Trade Structure

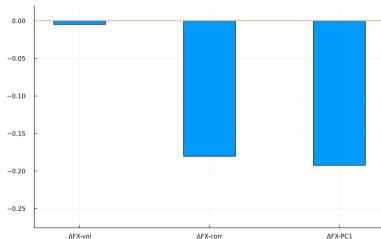
- 3×3 (export-biased, balanced, import-biased) $\times$ (random, baseline, perfect)



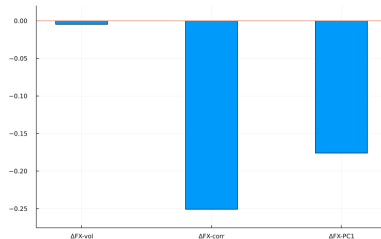
- Two-way traders with balanced import and export and perfect country matching can hedge most of exchange rate risk!

Counterfactual ER Environment: Currency Peg

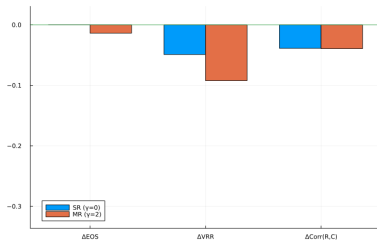
FX environment Δ (vs baseline) — USD peg ($\kappa=0.00$)



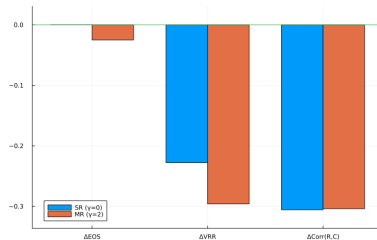
FX environment Δ (vs baseline) — Basket peg ($\kappa=0.30$)



Natural hedge Δ (vs baseline) — USD peg ($\kappa=0.00$)

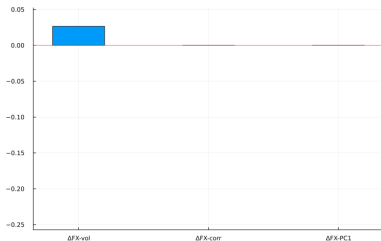


Natural hedge Δ (vs baseline) — Basket peg ($\kappa=0.30$)

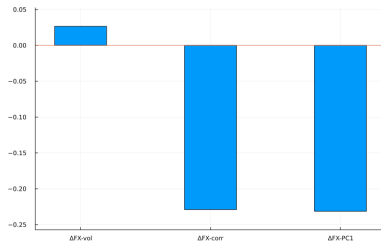


Counterfactual ER Environment: Volatility Amplification

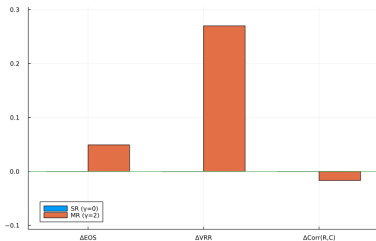
FX environment Δ (vs baseline) — Global volatility $\times 2.00$



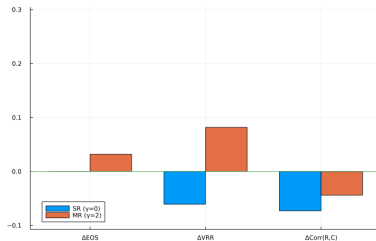
FX environment Δ (vs baseline) — Idiosyncratic volatility $\times 2.00$



Natural hedge Δ (vs baseline) — Global volatility $\times 2.00$



Natural hedge Δ (vs baseline) — Idiosyncratic volatility $\times 2.00$



Outline

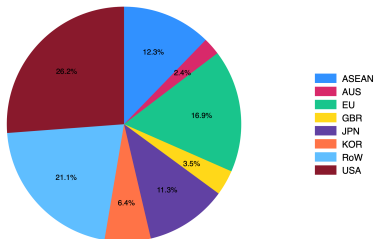
- 1 Introduction
- 2 Empirical Facts
- 3 Model
- 4 Numerical Analysis
- 5 Conclusion**

Conclusion

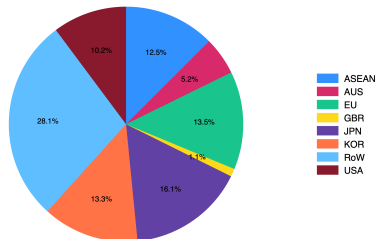
- This paper shows how **import-export linkages** shapes firms' responses to exchange rate uncertainty.
- I provide **new empirical evidence** showing that two-way traders with import-export alignment are less sensitive to exchange rates.
- I develop **a new trade model** with pricing-to-market + global sourcing + nominal price wedge from exchange rates.
- I highlight **a new strategy** for risk hedging → Firms face profit-variance trade-off under exchange rate risk can adjust its export-import alignment.
- The structural model could test hedging effectiveness in counterfactual initial trade structure and exchange rate environments.
- **Implications:** Why trade and financial policies interact in emerging markets? What governments can do to protect firms from global risk? What this means for supply chain resilience and diversification?

Motivating Facts: Top Markets and Origins

China's top export markets by value (2000-2015)



China's top import origins by value (2000-2015)



▶ back

Natural Hedging vs Financial Hedging

Summary

- In EMDEs, corporate use of FX derivatives is limited; instruments are short-tenor; access is constrained by actual-needs rules, documentation, and margin requirements.
- As a result, natural hedging (matching currency in operations and balance sheets) is often the only scalable and affordable risk-management channel for trade-exposed firms.

China: limited take-up and segmentation

- Corporate hedge ratios remain far below full coverage.
- Actual-needs requirements and CNY–CNH market segmentation create pricing and liquidity frictions.

Policy and access frictions across EMDEs

- India: documentation of underlying exposure, position limits, designated dealers.
- Indonesia: mandatory hedging ratios for firms with FX debt; liquidity and rating tests.
- Malaysia, Philippines, Thailand: restrictions on NDFs and non-resident positions; coterminous tenors with underlying.

Market depth, tenor, and transaction costs

- Wider bid–ask spreads in APAC EM FX and limited availability beyond one year.
- Shorter available tenors relative to major currencies.

Funding frictions and the CIP basis

- Persistent deviations from CIP in EMDEs increase hedge costs, especially in stress.

Collateral and operational barriers

- ISDA/CSA documentation, daily margining, and collateral needs tie up liquidity.
- Negotiation and onboarding timelines are lengthy, posing barriers for SMEs.

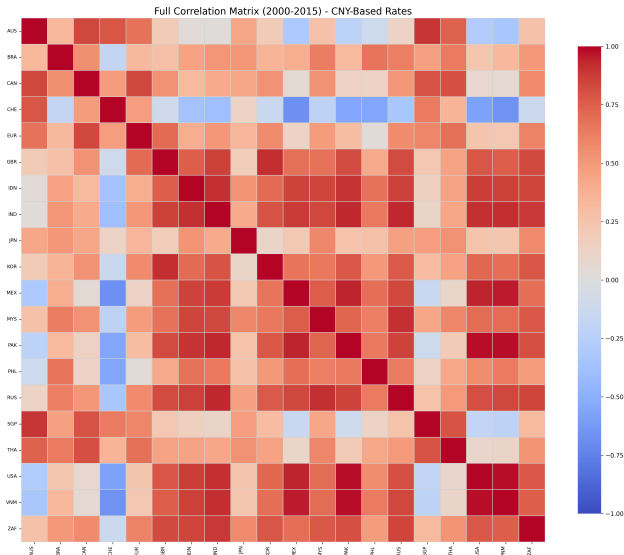
Case beyond Asia

- Public and MDB-supported programs to extend longer-dated and cheaper hedges indicate market scarcity at long maturities.

Implications

- For EMDE exporters and importers, financial hedges are often incomplete and costly.
- Natural hedging now becomes the ideal substituting approach under binding frictions.

Full Estimated Covariance Matrix



Parametrization

Table: Parametrization

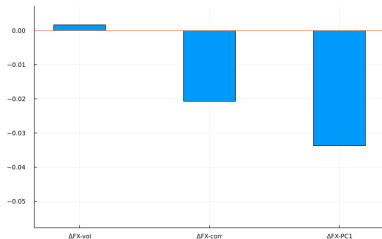
Parameter	Symbol	Value	Method & Source
Demand elasticity	σ	3.8	Oberfield and Raval (2021)
Output elasticity of materials	α	0.6	Oberfield and Raval (2021)
PCP share	λ	0.95	Li, Ma and Xu (2015), Li, Lu and Zhao (2025)
Elasticity of substitution between domestic and foreign inputs	ϵ	4.0	Halpern, Koren and Szeidl (2015), Gopinath and Neiman (2014)
Elasticity of substitution between inputs from different origins	κ	3.0	Feenstra et al. (2018)
Risk aversion	γ	2.0 - 5.0	Chetty (2006)

Note: This table lists the parameter values assigned from literature and calibration.

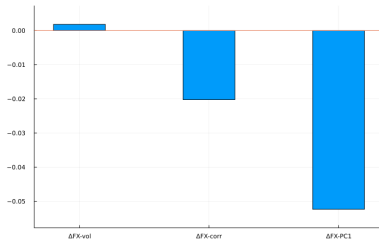
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Counterfactual ER Environment: Jump-mix Risk

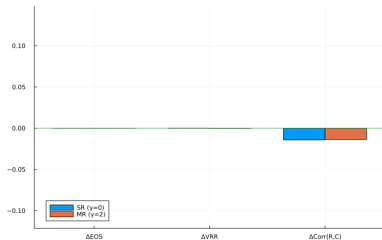
FX environment Δ (vs baseline) — Jump risk (USD): $p=10\%$, $J=20\%$



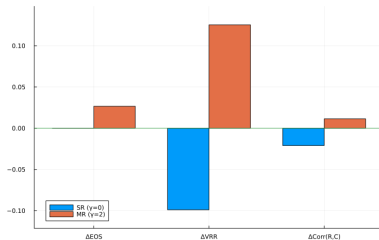
FX environment Δ (vs baseline) — Jump risk (EUR): $p=10\%$, $J=20\%$



Natural hedge Δ (vs baseline) — Jump risk (USD): $p=10\%$, $J=20\%$



Natural hedge Δ (vs baseline) — Jump risk (EUR): $p=10\%$, $J=20\%$



[back](#)

References I

- Alfaro, Laura, Mauricio Calani, and Liliana Varela.** 2021. "Granular corporate hedging under dominant currency." National Bureau of Economic Research.
- Allayannis, George, and Eli Ofek.** 2001. "Exchange rate exposure, hedging, and the use of foreign currency derivatives." *Journal of international money and finance*, 20(2): 273–296.
- Allayannis, George, and James P Weston.** 2001. "The use of foreign currency derivatives and firm market value." *The review of financial studies*, 14(1): 243–276.
- Amiti, Mary, Oleg Itskhoki, and Jozef Konings.** 2014. "Importers, exporters, and exchange rate disconnect." *American Economic Review*, 104(7): 1942–1978.
- Blaum, Joaquin.** 2024. "Global firms in large devaluations." *The Quarterly Journal of Economics*, 139(4): 2427–2474.
- Chetty, Raj.** 2006. "A new method of estimating risk aversion." *American Economic Review*, 96(5): 1821–1834.
- Devereux, Michael B, Wei Dong, and Ben Tomlin.** 2017. "Importers and exporters in exchange rate pass-through and currency invoicing." *Journal of International Economics*, 105: 187–204.
- Fauceglia, Dario, Anirudh Shingal, and Martin Wermelinger.** 2014. "Natural hedging of exchange rate risk: the role of imported input prices." *Swiss Journal of Economics and Statistics*, 150: 261–296.
- Feenstra, Robert C, Philip Luck, Maurice Obstfeld, and Katheryn N Russ.** 2018. "In search of the Armington elasticity." *Review of Economics and Statistics*, 100(1): 135–150.

References II

- Gopinath, Gita, and Brent Neiman.** 2014. "Trade adjustment and productivity in large crises." *American Economic Review*, 104(3): 793–831.
- Gopinath, Gita, and Roberto Rigobon.** 2008. "Sticky borders." *The Quarterly Journal of Economics*, 123(2): 531–575.
- Gopinath, Gita, Oleg Itskhoki, and Roberto Rigobon.** 2010. "Currency choice and exchange rate pass-through." *American Economic Review*, 100(1): 304–336.
- Halpern, László, Miklós Koren, and Adam Szeidl.** 2015. "Imported inputs and productivity." *American economic review*, 105(12): 3660–3703.
- Li, Hongbin, Hong Ma, and Yuan Xu.** 2015. "How do exchange rate movements affect Chinese exports?—A firm-level investigation." *Journal of International Economics*, 97(1): 148–161.
- Li, Yao Amber, Lingfei Lu, and Tengyu Zhao.** 2025. "Exchange rate pass-through and importers' credit constraints: Evidence from China." *Journal of Economic Behavior & Organization*, 236: 107044.
- Li, Yao Amber, Sichuang Xu, Stephen R Yeaple, and Tengyu Zhao.** 2024. "Bilateral Economies of Scope." *Review of Economics and Statistics*, 1–45.
- Oberfield, Ezra, and Devesh Raval.** 2021. "Micro data and macro technology." *Econometrica*, 89(2): 703–732.
- Raddatz, Claudio E.** 2011. "Over the hedge: exchange rate volatility, commodity price correlations, and the structure of trade." World Bank Policy Research Working Paper 5590.
- Salomao, Juliana, and Liliana Varela.** 2022. "Exchange rate exposure and firm dynamics." *The Review of Economic Studies*, 89(1): 481–514.